

Flow of Program - Flowchart & Pseudocode

Flowchart

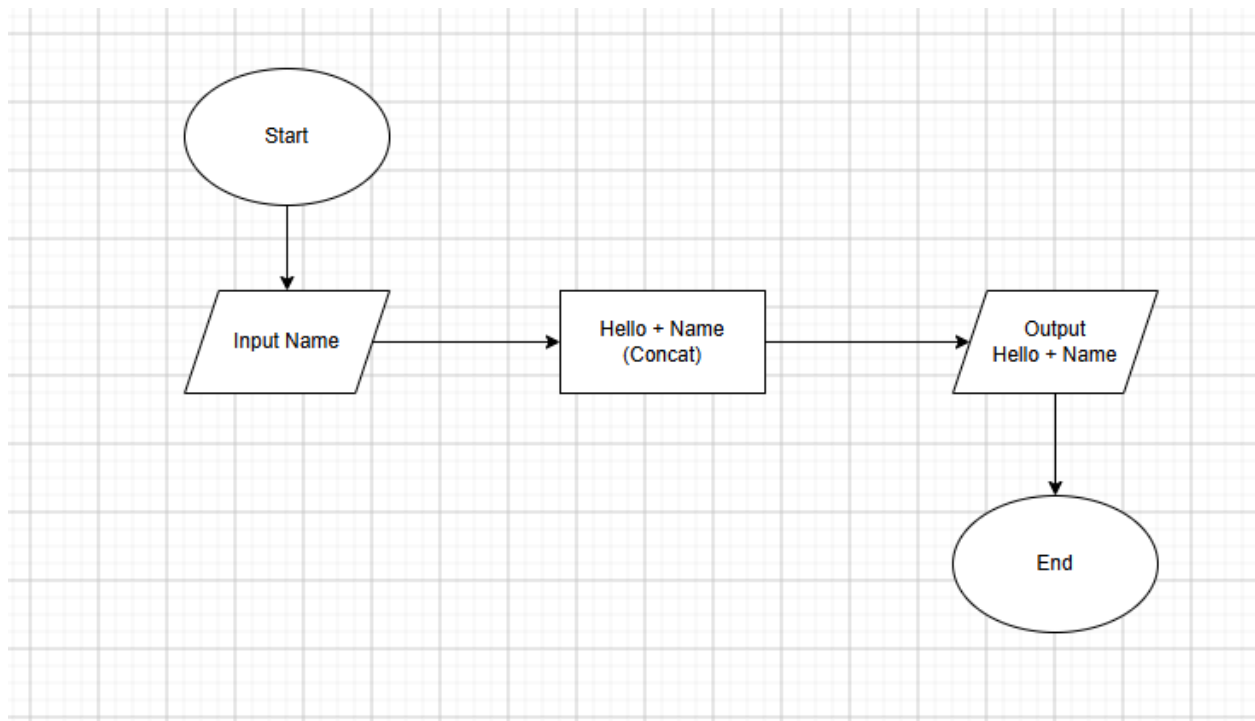
- Flow chart uses the following symbols for representing various functions

Is represented	By
Start / Stop	Ellipse Or Oval
Input / Output	Parallelogram
Processing	Rectangle
Condition	Rhombus Or Diamond

- Used to visualize our thought process
- Sometimes used to describe complex architecture

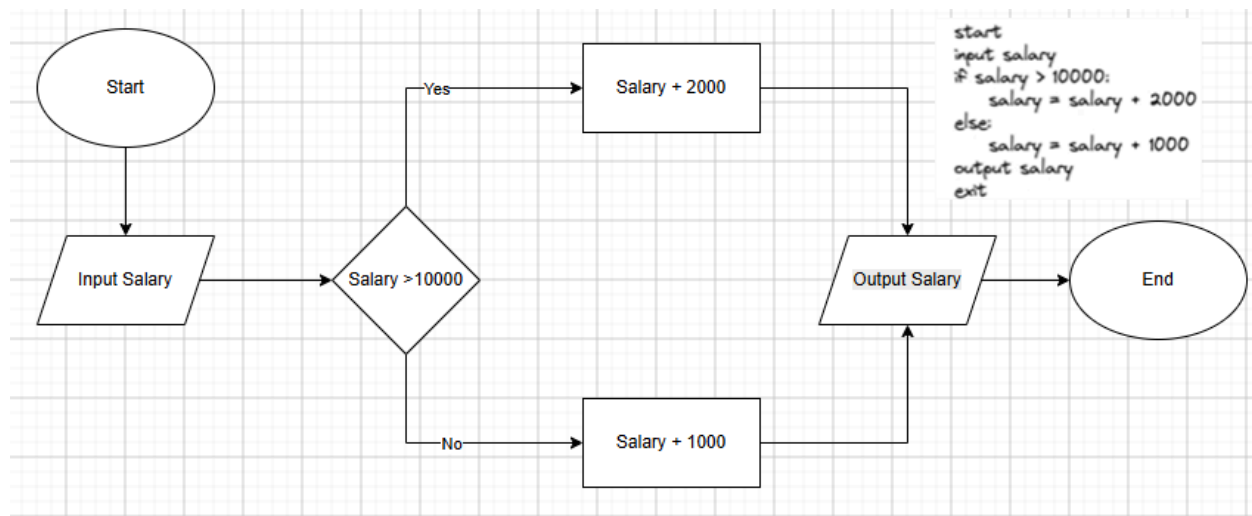
Flow Exercise - 1:

Take a name and output hello name



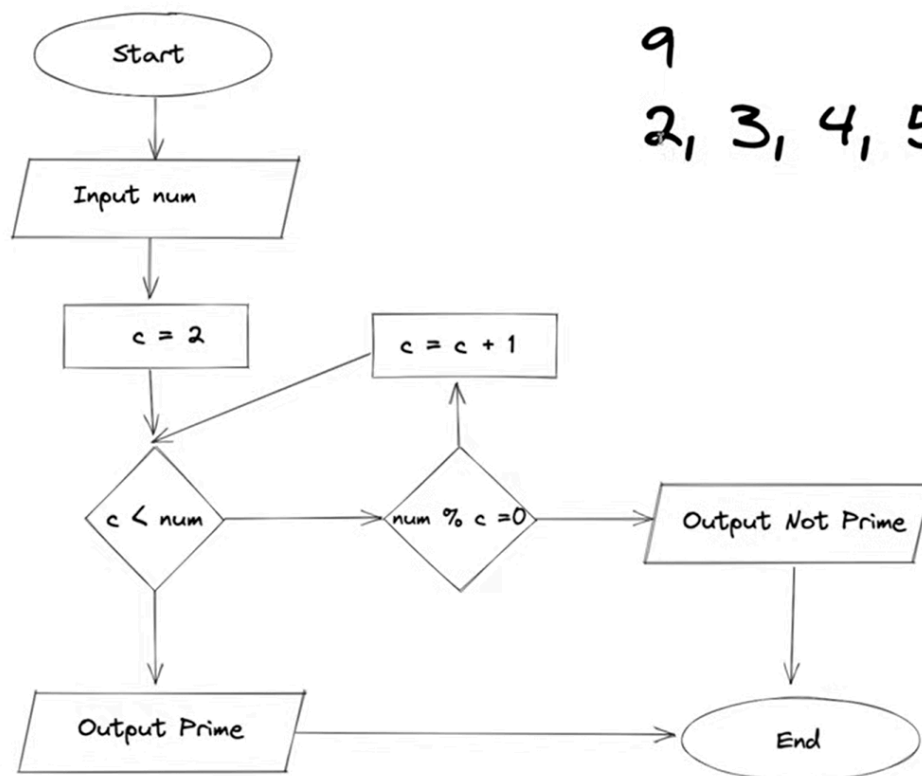
Flow Exercise - 2:

Take input of a salary and if the salary is greater than 10,000 then add a bonus of 2000, Otherwise add a bonus of 1000



Flow Exercise - 3:

Input a number check if it is prime or not



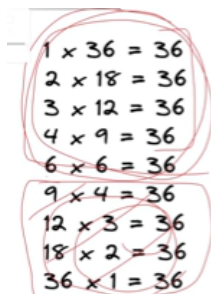
Pseudocode

- If we want to share algorithms with someone we share it with them in the form of pseudocode. Programming language doesn't matter. It captures the essence of what an algorithm does.
- Basically a rough code.
- Looks as shown below

```
start
input num
c = 2
while c < num:
    if num % c == 0:
        output "not prime"
        exit
    c = c + 1
end while
output "prime"
exit
```



- Tab space is used after the while condition to represent what is running after while is running when the condition specified inside the loop is satisfied.
- So, $c < \text{num}$ should be true and the if statement should check its condition while $c < \text{num}$ is true.
- Here in the prime not prime program, if the number is large, While checking the condition, the number repeats itself as shown below.



1 x 36 = 36
2 x 18 = 36
3 x 12 = 36
4 x 9 = 36
6 x 6 = 36
9 x 4 = 36
12 x 3 = 36
18 x 2 = 36
36 x 1 = 36

- So, while checking 36 as prime or not, if we check the divisibility only up to 6 that would be fine as other cases repeat themselves.
- So, we need to only check up to the square root of the number. Here, $\sqrt{36} = 6$. So, we need to check only up to 6.
- This would reduce the complexity of the program by its square root. So, in the above image, it should be $\sqrt{\text{num}}$.